

Lecture: Distributed Systems and Consensus Architecture.

Key Concepts:

1. The CAP Theorem states that a distributed data store can only guarantee two out of three qualities: Consistency, Availability, and Partition Tolerance.
2. Raft Consensus Algorithm elects a designated leader who manages log replication across follower nodes.
3. Two-Phase Commit ensures atomic transaction commits across distributed database shards.
4. Vector clocks allow tracking causal ordering of asynchronous events without synchronized physical clocks.
5. Eventual consistency enables replicas to converge on identical state given sufficient latency.